Department of Physics
Sharif University of Technology
Cosmology Group Weekly Seminars

دانشکده فیزیک دانشگاه صنعتی شریف

سمینارهای هفتگی گروه کیهان شناسی

لطفا به زمان و مکان سمینار توجه فرمایید

سمینار به صورت ترکیبی (هیبرید) برگزار می شود

Cosmological Signatures of Curvature-Coupled Dark Energy

Abdolali Banihashemi

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Abstract:

In this talk, I will present a curvature-coupled dark energy model within the framework of scalar tensor theories, where a quintessence field is nonminimally coupled to the Ricci scalar. The model is characterized by a shifted quartic coupling, $f(\varphi) = \alpha(\varphi^2 - \varphi^2_today)^2$, and an inverse power-law potential, $V(\varphi) = \Lambda\varphi^{-}\sigma$
I will show how this specific coupling allows an effective crossing of the phantom divide at late times while satisfying local gravity constraints. At the same time, the scalar field can modify the expansion history before recombination, potentially alleviating the H_0 tension. Using a modified version of hi_class, I will discuss the model's predictions for the cosmological background and linear perturbations, highlighting its characteristic signatures in matter clustering, weak lensing, and the Integrated Sachs-Wolfe effect. Finally, I will discuss the prospects for further investigation, including a full parameter space analysis and nonlinear N-body simulations.

یکشنبه ۲۴ آبان ۱۴۰۵، ساعت ۱۸:۱۵

Sunday 15 November 2026 – 18:15 Tehran Time

Hybrid Seminar

۱۸:۰۰ - ۱۸:۱۵ | پذیرایی ساده و گفتگو

https://vc.sharif.edu/ch/cosmology

Physics Department fifth floor –Room 512 / دانشکده فیزیک – طبقه ۵ – اتاق ۵۱۲

Enter as a Guest – گزینه ورود به صورت مهمان

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